

Smart Distributed Acoustic Calibration System (SDACS)

Engineering Design Presentation

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Bachelor of Applied Engineering – Electronic Systems Engineering

Capstone I

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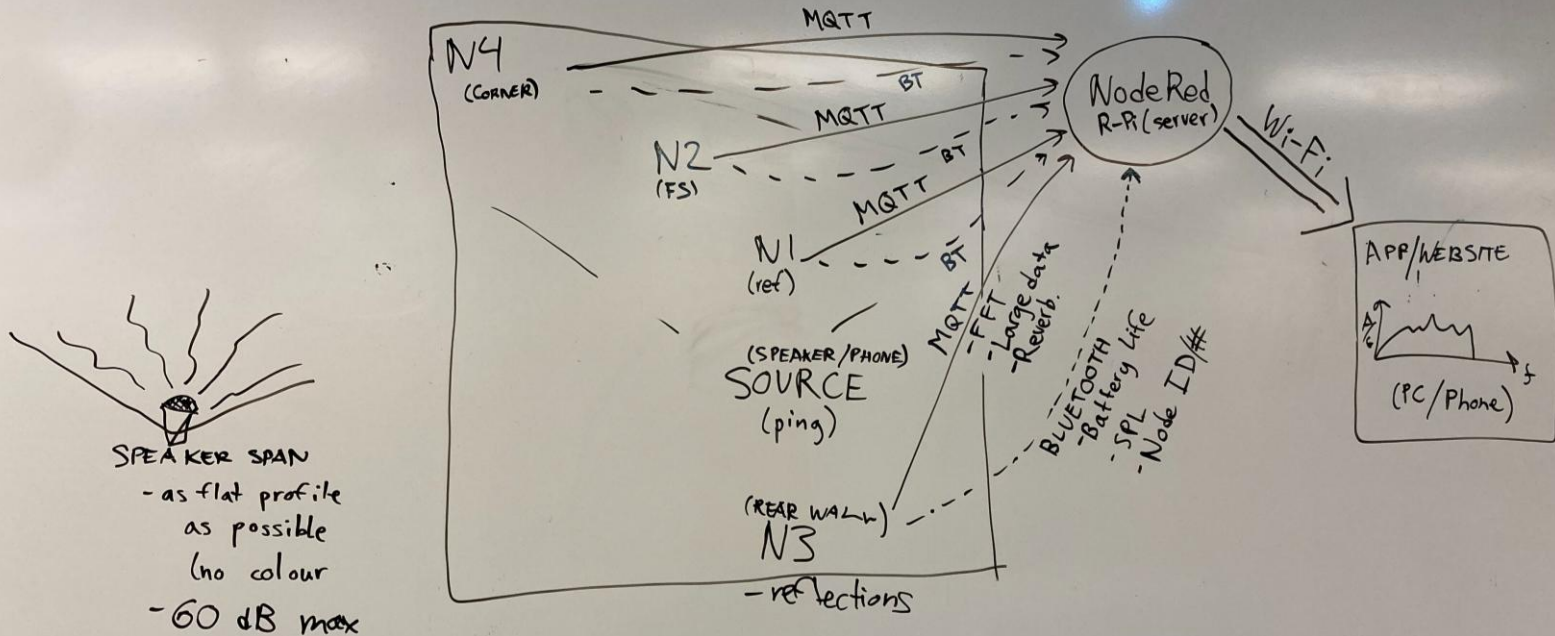
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01 INTRODUCTION



01 INTRODUCTION – Who We Are



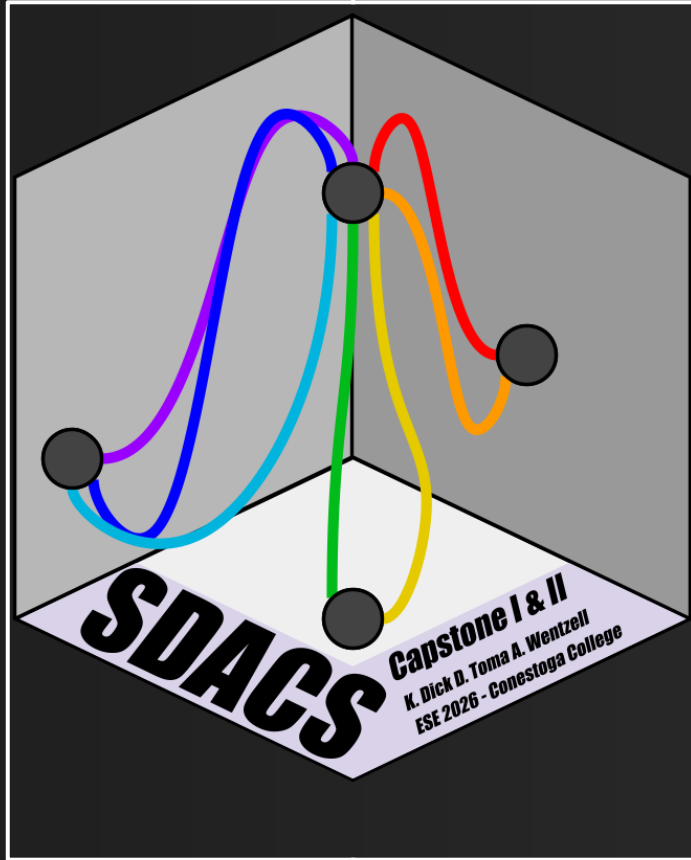
DAVID TOMA
Hardware Lead
ESE 2026



KYLE DICK
Firmware Lead
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AMY WENTZELL
Software Lead
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01 INTRODUCTION - SDACS

Smart

Distributed

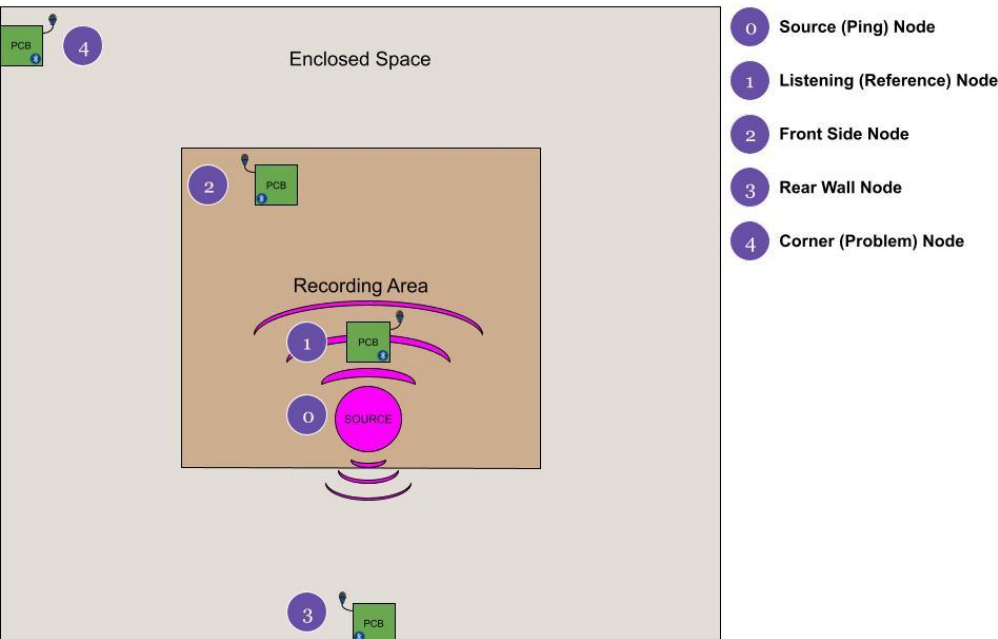
Acoustic

Calibration

System

01 INTRODUCTION - SDACS

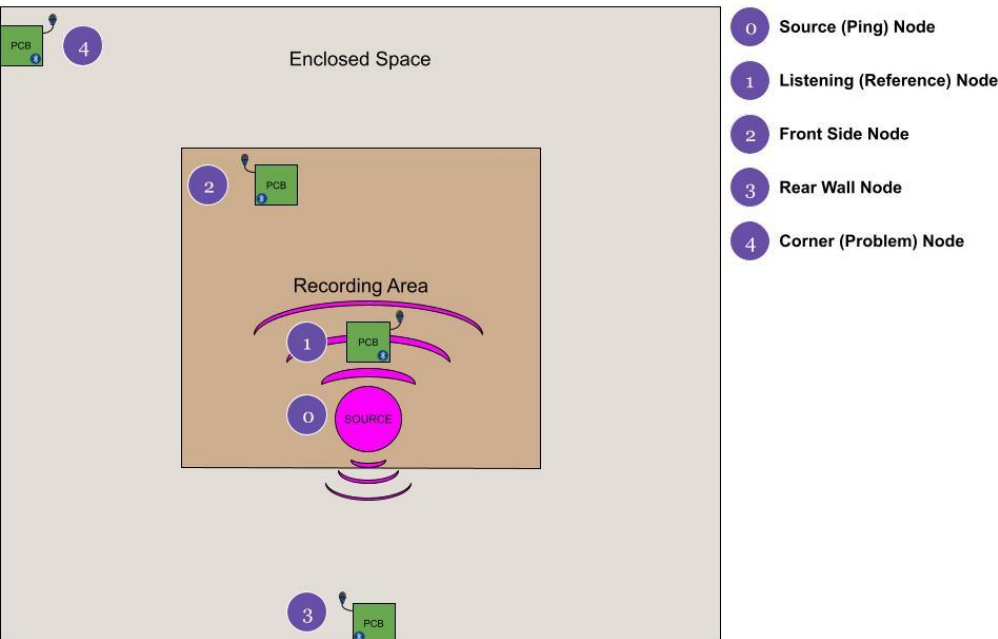
Smart Distributed Acoustic Calibration System - Conceptual Diagram



1. Synchronize the nodes on the network.
2. Determine positioning of nodes relative to the server.

01 INTRODUCTION - SDACS

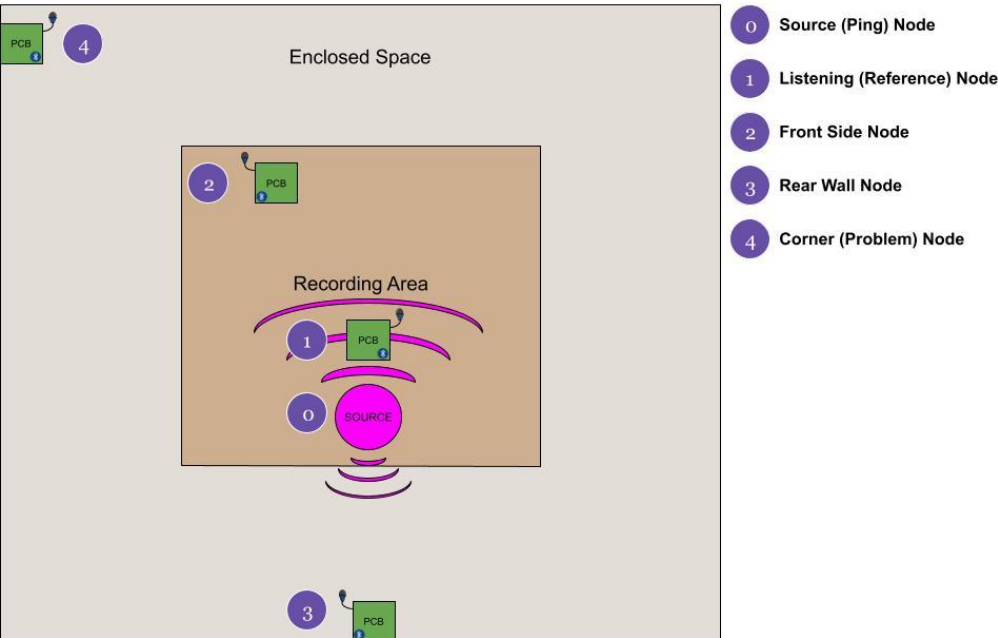
Smart Distributed Acoustic Calibration System - Conceptual Diagram



3. Perform the frequency sweep.
Record the audio on the recording nodes.
4. Calculate FFTs, SPLs, and reverberation times (RT60) at node(s).
5. Send results from the recording nodes to the server.

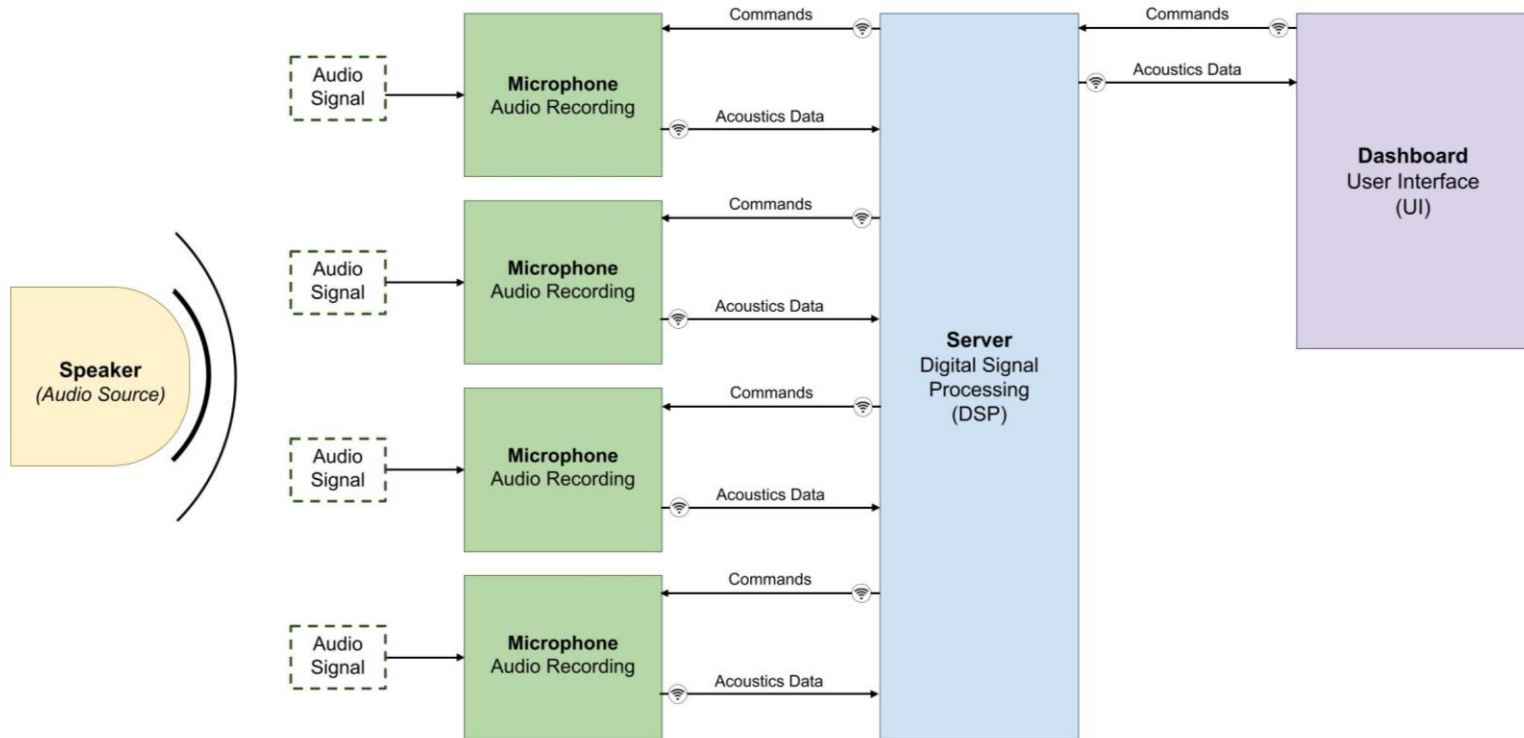
01 INTRODUCTION - SDACS

Smart Distributed Acoustic Calibration System - Conceptual Diagram



6. Create three-dimensional model of the indoor spatial acoustics using the acquired data.
7. Send results from the server to the client.
8. Display the model(s), results to the client using the website / application.

02 SYSTEM REQUIREMENTS



02 SYSTEM REQUIREMENTS

MUST

Priority	Requirement
MUST	Enable simultaneous acoustic measurements from multiple indoor locations.
MUST	Utilize ESP32-based nodes with digital MEMS microphones for reliable audio capture.
MUST	Support coordinated multi-node recording for spatial comparison.
MUST	Capture, buffer, and wirelessly transmit audio data to a Raspberry Pi 5 for centralized processing.
MUST	Perform frequency-domain (FFT-based) analysis on recorded data.
MUST	Provide a web-based dashboard to display measurement results.
MUST	Demonstrate a functional prototype in a representative indoor environment.
MUST	Execute structured verification and validation testing to confirm system performance.

02 SYSTEM REQUIREMENTS

SHOULD, COULD, WON'T

Priority	Requirement
SHOULD	Allow remote configuration and control of measurement nodes.
SHOULD	Store measurement data for post-processing and session comparison.
SHOULD	Implement automated acoustic metrics such as RT60 estimation.
COULD	Include environmental sensing (e.g., temperature, humidity).
COULD	Perform limited on-node signal processing to reduce data transmission load.
COULD	Support scalability to additional nodes and third-party tool integration.
WON'T	Perform professional audio production or replace commercial acoustic systems.
WON'T	Include active acoustic treatment hardware or support industrial-scale deployment.

03 SYSTEM DESIGN

Firmware



ESP-IDF
VS Code
GitHub



Real-time I2S
Capture @48 kHz

Chunk-based
buffering

/sdcard



MQTT

Server Side

(Node-RED)



Raspberry Pi 5 Broker
(Server)



MQTT Sub to Topics

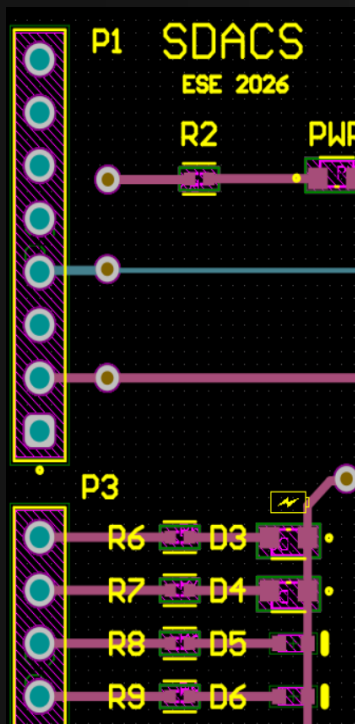
Features
Heartbeat
Battery

MQTT

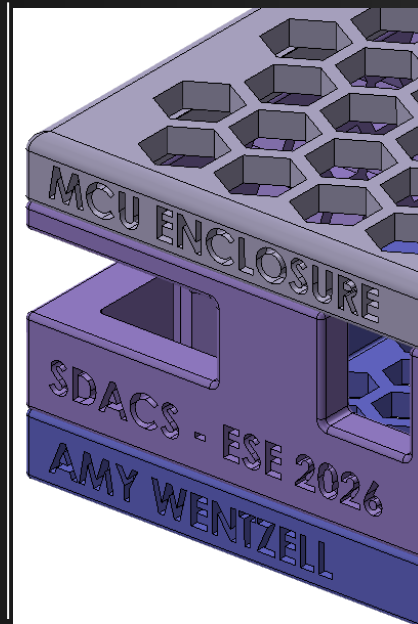
CSV & SQLite
Export



Hardware



Enclosure



Firmware (Data Streaming)

I2S + I2C Acquisition

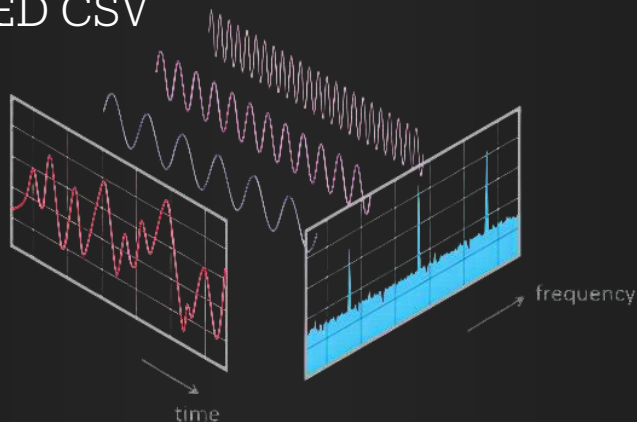
- I2S MEMS mic configured at 48 kHz to capture full audible range
- Provides sufficient resolution for FFT computation later in MATLAB
- I2C T/H sensor sampled during capture
- I2C Fuel Gauge track battery
- Heartbeat tracking Wi-Fi connection



Firmware (Audio Feature Extraction)

Key Acoustic Features

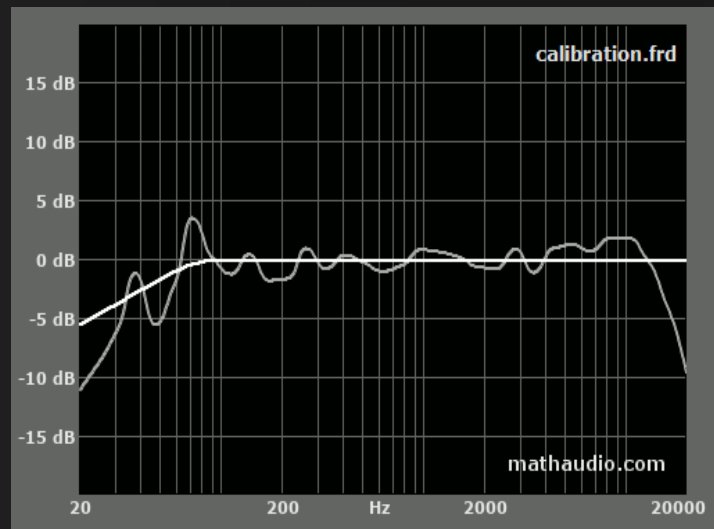
- RMS $\rightarrow$ signal energy
- dbFS $\rightarrow$ decibels interpreted at the MCU (voltages)
- FFT peak $\rightarrow$ esp_dsp
- Logged in SD card & Node-RED CSV



Chunk Based Buffering

No large buffers (less SRAM)

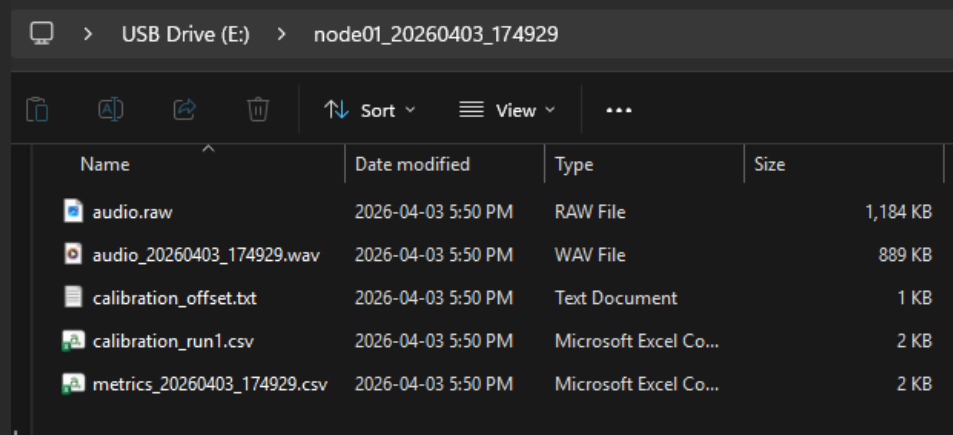
Fixed chunks $\sim 10\text{ms}$ (512 samples
@ 48kHz $\sim 10.7\text{ms}$)



Firmware (Local Data Capture)

SD Card Logging Ensuring System Reliability and Validation

- Provides local storage of raw and processed data (.csv, .wav, .raw)
- Validation against data published to Node-RED (.csv)
- Includes time-stamped T/H & Battery data
- Backup if Wi-fi fails
- Pipeline for analysis in MATLAB



Name	Date modified	Type	Size
audio.raw	2026-04-03 5:50 PM	RAW File	1,184 KB
audio_20260403_174929.wav	2026-04-03 5:50 PM	WAV File	889 KB
calibration_offset.txt	2026-04-03 5:50 PM	Text Document	1 KB
calibration_run1.csv	2026-04-03 5:50 PM	Microsoft Excel Co...	2 KB
metrics_20260403_174929.csv	2026-04-03 5:50 PM	Microsoft Excel Co...	2 KB



ESP32 → Capture Task → SD Card (.csv/.wav/.raw)



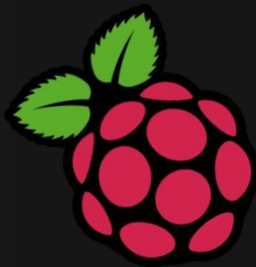
MQTT (parallel stream) → Node-RED (.csv / SQLite)

Firmware (MQTT-Based Comms.)

MIC / TH / FG → ESP32 → Firmware → MQTT → Node-RED → Dashboard

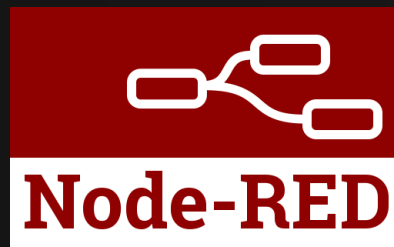
WHAT:

- Lightweight pub/subs model
- Pub/sub to topics:
(*sdacs/node/node01/features*)
- Designed for IoT and Embedded Systems
- Supports multiple nodes & topics (ideal for SDACS)



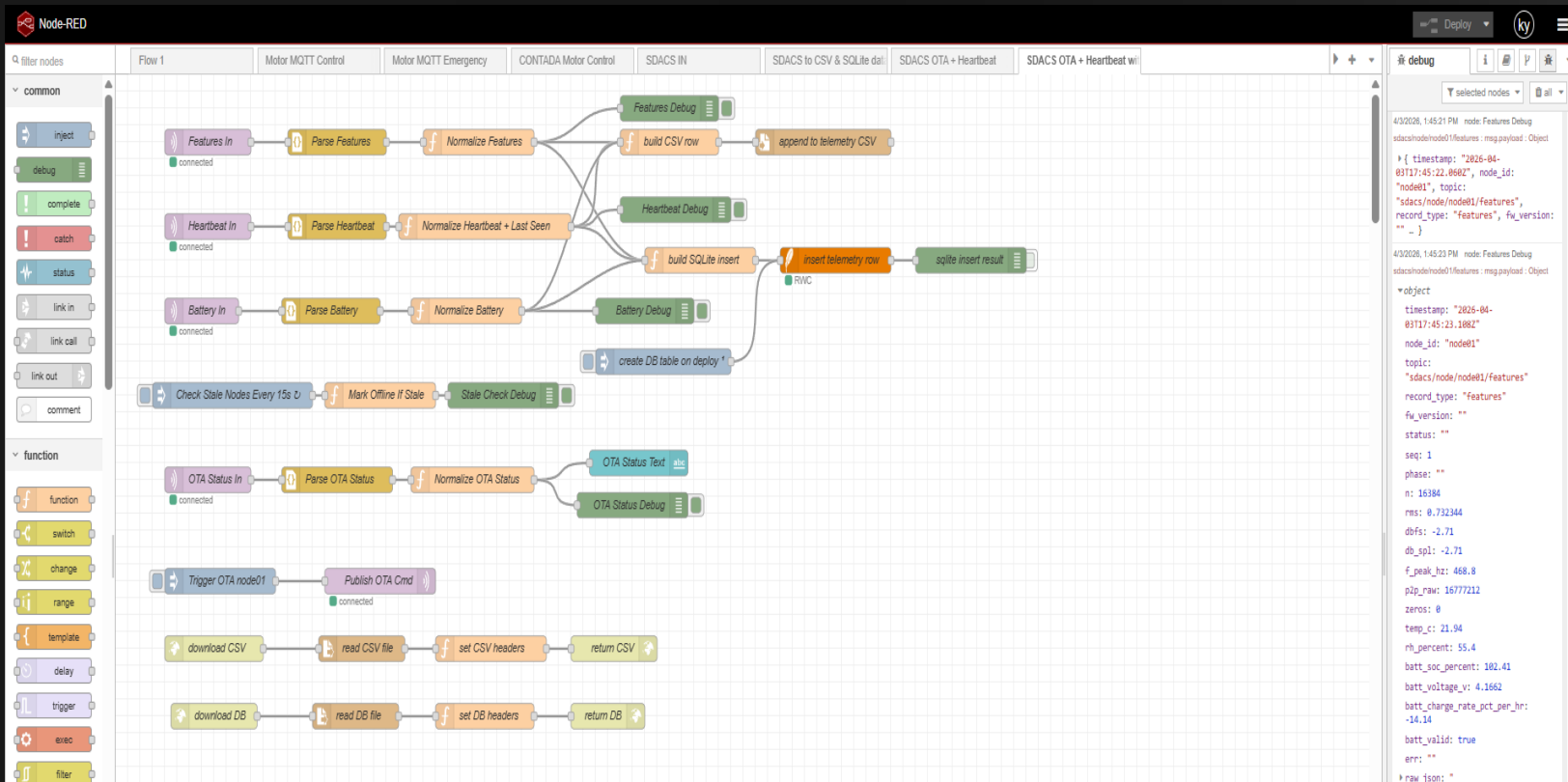
WHY:

- Enables multi-node deployment
- Integration with Node-RED
- Easy setup on Raspberry Pi Server (edge aggregating broker)



SDACS Node-RED Flow

ESP32s → MQTT Broker (RPi5) → Node-RED (PC) → CSV / SQLite → Dashboard / UI



03 SYSTEM DESIGN: MICROCONTROLLER

Why we chose it:

- **High Processing Capability**
Dual-core MCU (up to 240 MHz) handles real-time audio (I²S) and sensor data (I²C) simultaneously
- **Built-in Peripheral Support**
Native I²C, I²S, and GPIO eliminates need for extra interface hardware
- **Integrated Wireless Connectivity**
Onboard Wi-Fi and Bluetooth enables future IoT and remote monitoring features
- **Strong Memory Capabilities**
Sufficient SRAM and external flash support for buffering audio data and handling sensor processing efficiently



03

SYSTEM DESIGN: BATTERY

Why we chose it:

EEMB 3.7V 1100mAh 603449

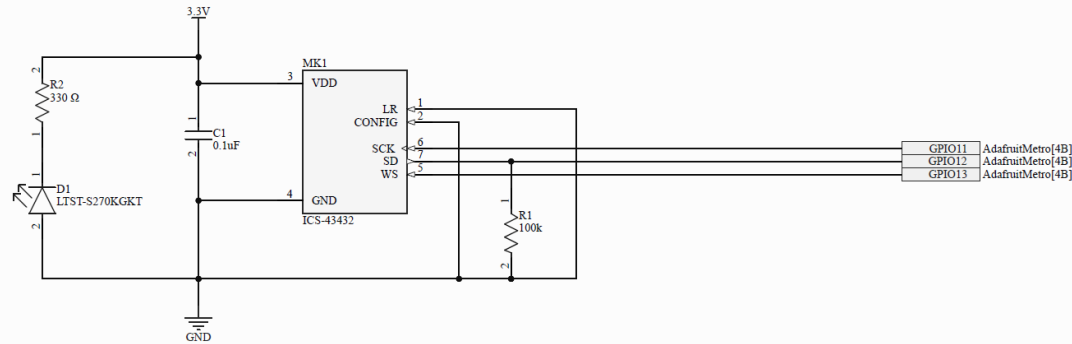
- **Suitable Voltage for our System**
 - 3.7 nominal output is ideal for powering the microcontroller
- **Adequate Capacity (1100 mAh)**
 - Provides sufficient runtime for continuous sensor monitoring and data acquisition
- **Compact and Lightweight**
 - Slim form factor (603449) fits easily within the PCB/enclosure design constraints
- **Rechargeable & Efficient**
 - Lithium polymer chemistry offers high energy density and supports repeated charging cycles



03 SYSTEM DESIGN

MEMS Mic Circuit (Mono)

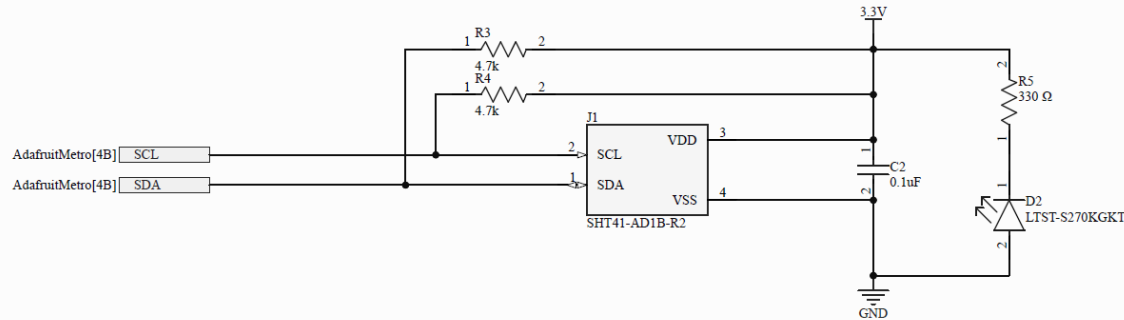
MEMS Microphone Circuit (Mono)



03 SYSTEM DESIGN

Temperature/Humidity Sensor

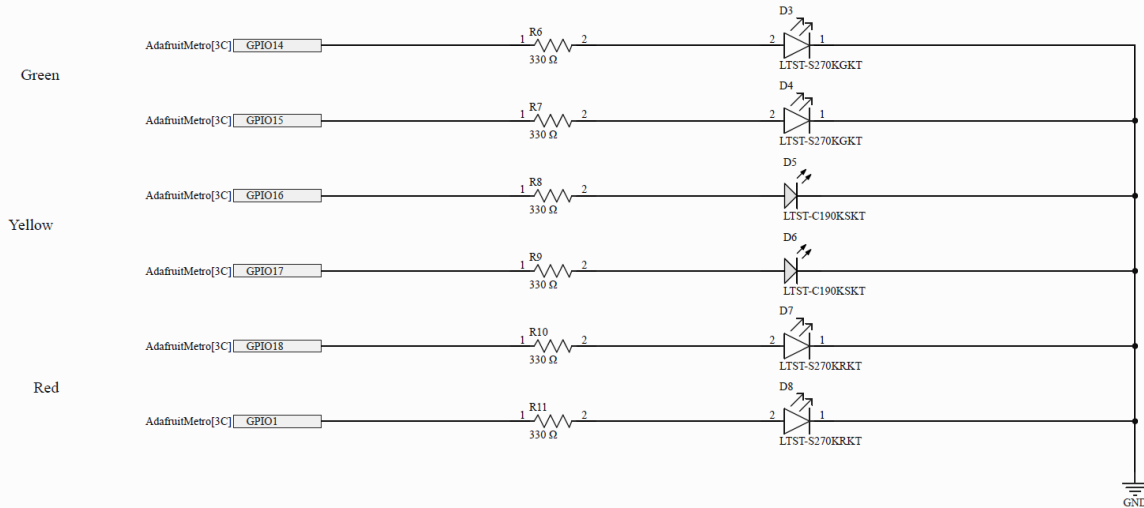
Temperature/Humidity Sensor



03 SYSTEM DESIGN

Power LEDs (Battery Life)

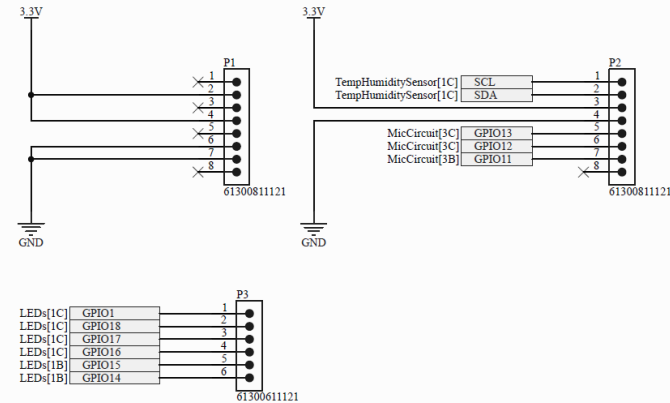
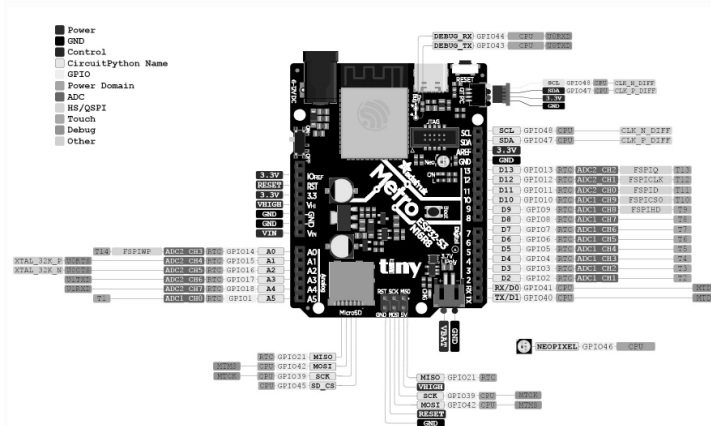
Power LEDs (Battery Life)



03 SYSTEM DESIGN

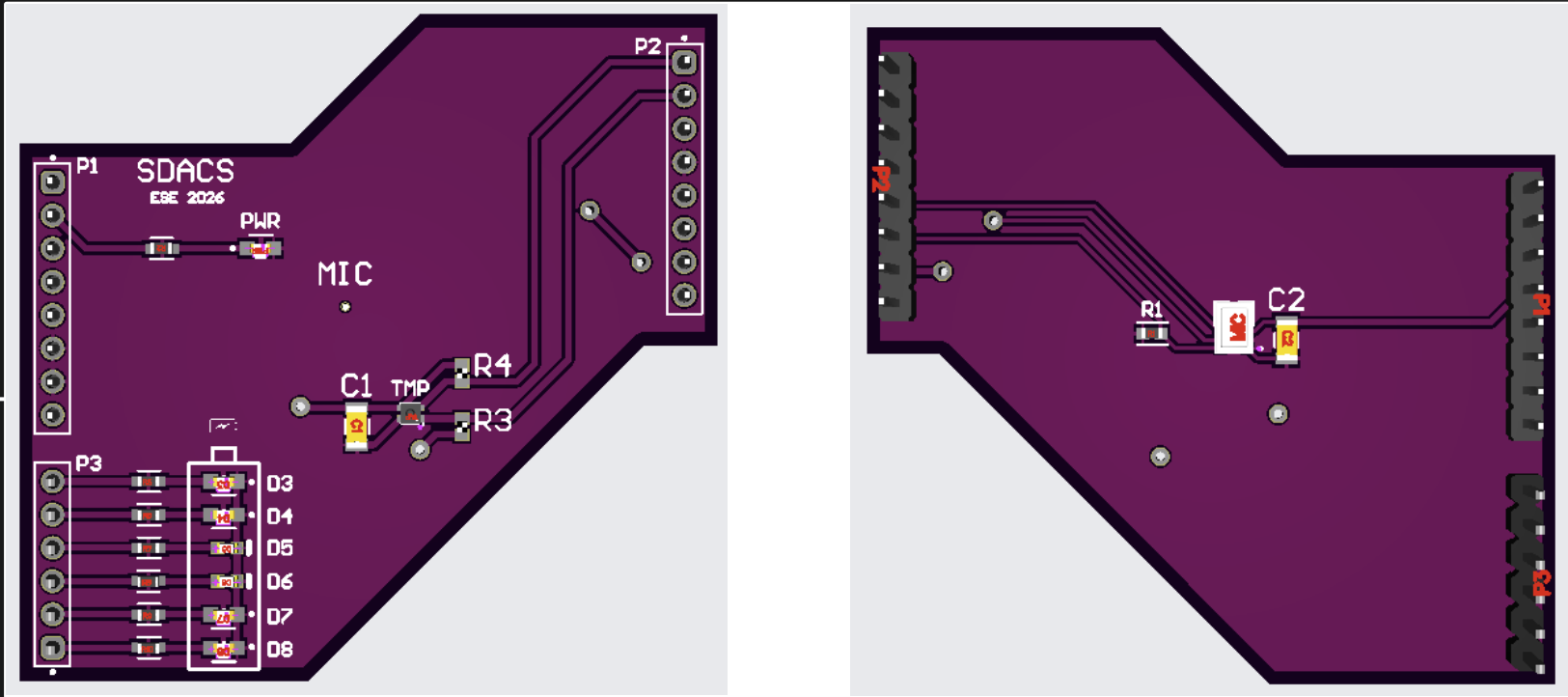
Metro Board Pinout (Headers)

Metro Board Pinout (Headers)



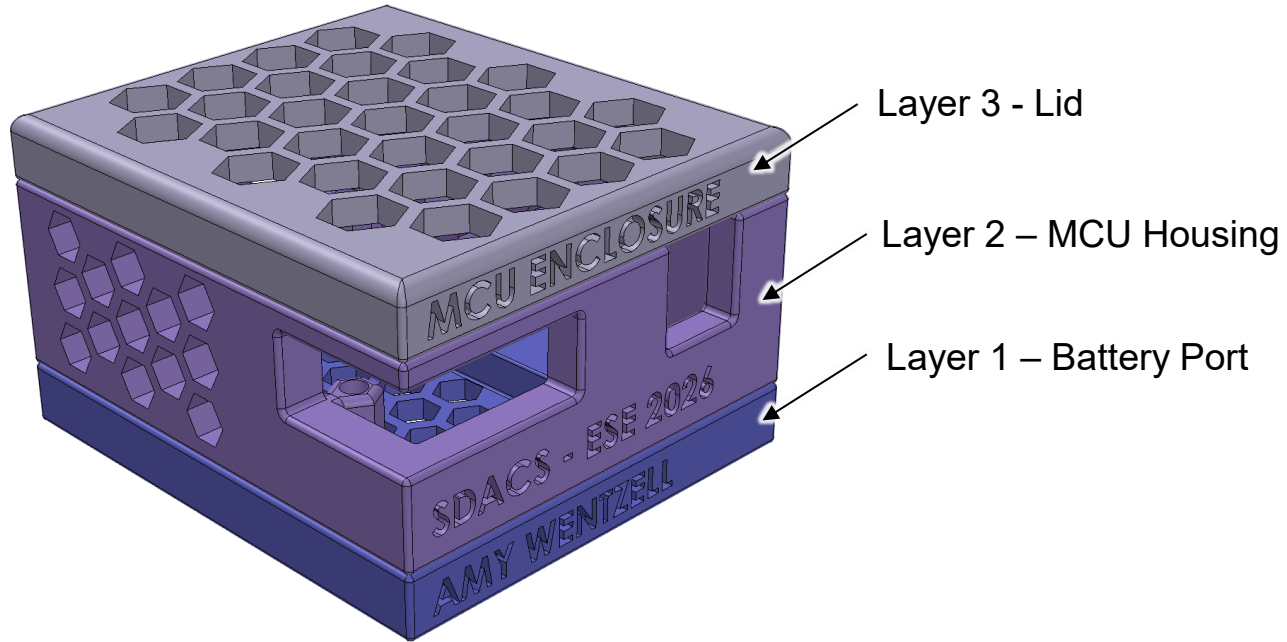
03 SYSTEM DESIGN

PCB Design



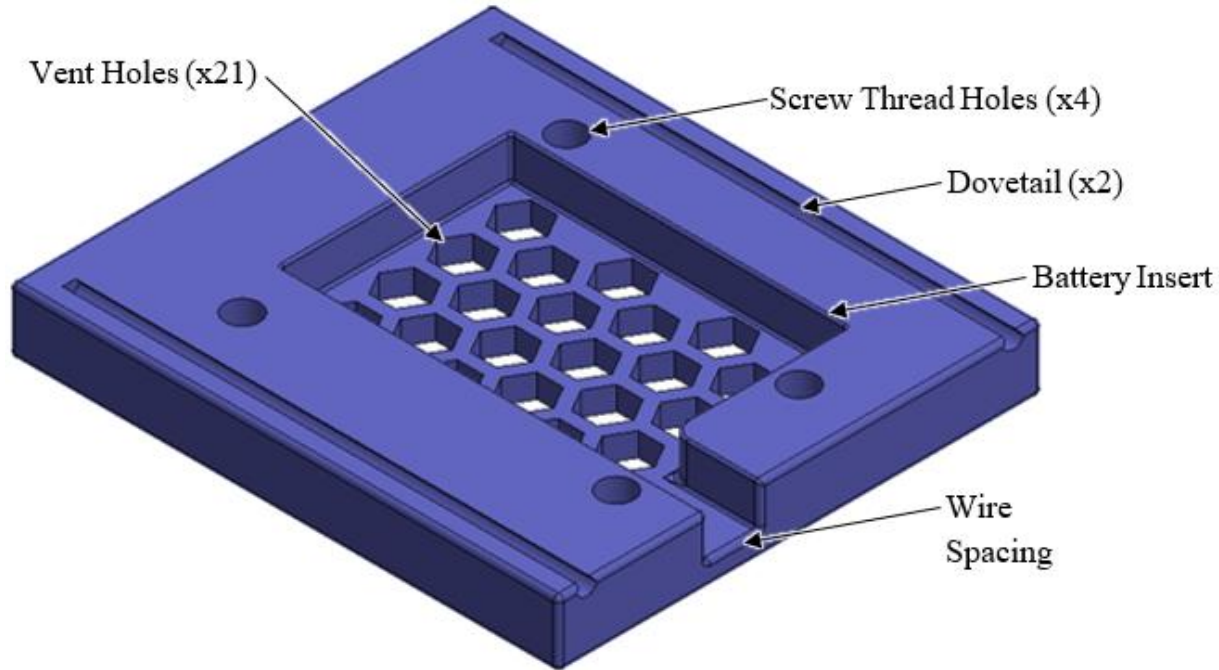
03 SYSTEM DESIGN

ENCLOSURE (Rev C)



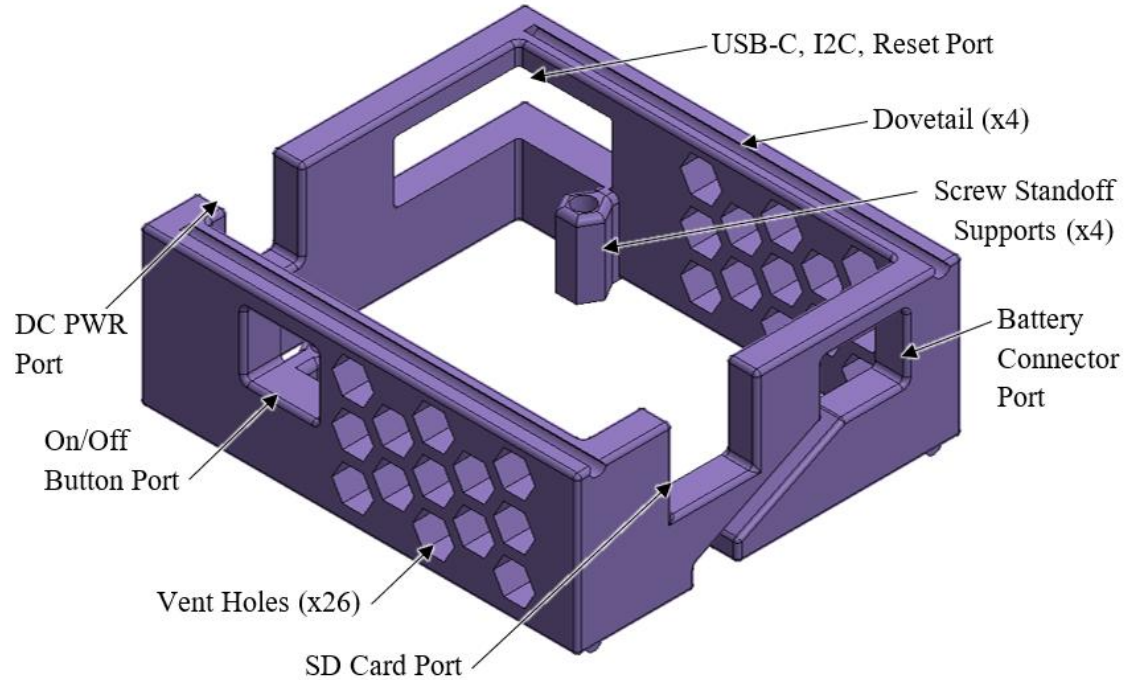
03 SYSTEM DESIGN

BATTERY PORT (Rev C)



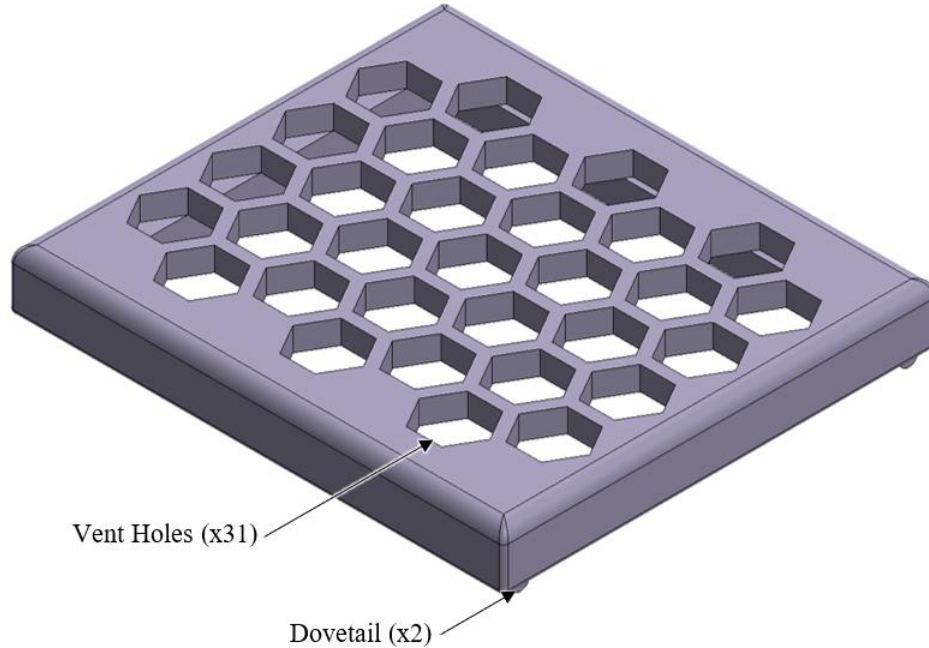
03 SYSTEM DESIGN

MCU HOUSING (Rev C)



03 SYSTEM DESIGN

LID (Rev C)

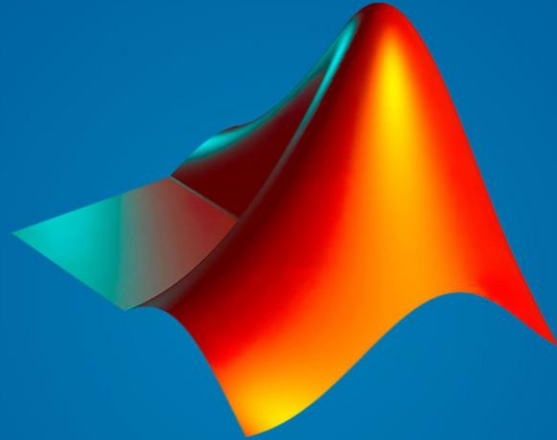


04

MAJOR UNITS UNDER DESIGN AND DEVELOPMENT



Flutter

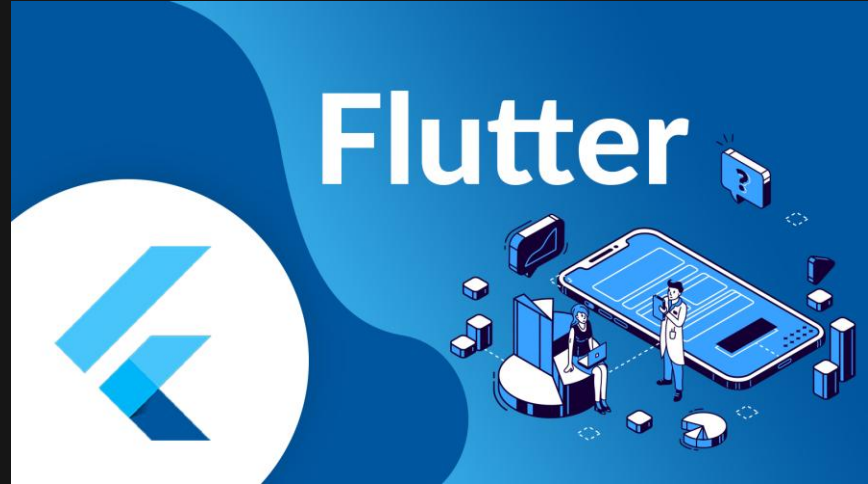


MATLAB

04 UNDER DEVELOPMENT

Flutter App Development

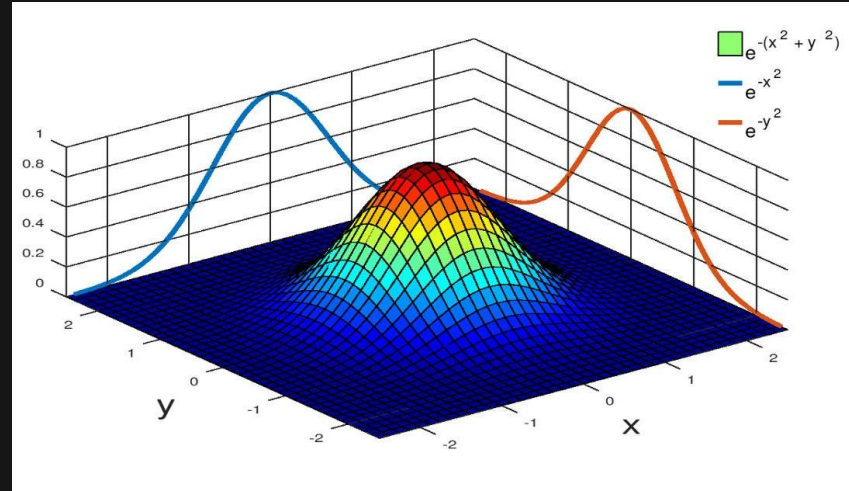
- App development for remote to calibrate nodes and display features
- Create a clean UI for all the sensor outputs
- 3D data visualization display for the user
- Wireless data integration: Connect app to the nodes via Bluetooth and Wi-fi



04 UNDER DEVELOPMENT

MATLAB Analysis / 3-D Modelling

- Node-RED integration
- Graphical representations of data
- Numerous libraries available for DSP, graphs, etc.
 - FSDA Toolbox
 - Gaussian Distribution



05

RISK ANALYSIS AND RISK MANAGEMENT PLAN



05 RISK ANALYSIS



TECHNICAL RISKS

Competitive Landscape, Node
Synchronization, Custom PCB
Complexity



PROJECT MANAGEMENT RISKS

Scope Creep, Time
Constraints



ORGANIZATIONAL RISKS

App and Website
Development



EXTERNAL RISKS

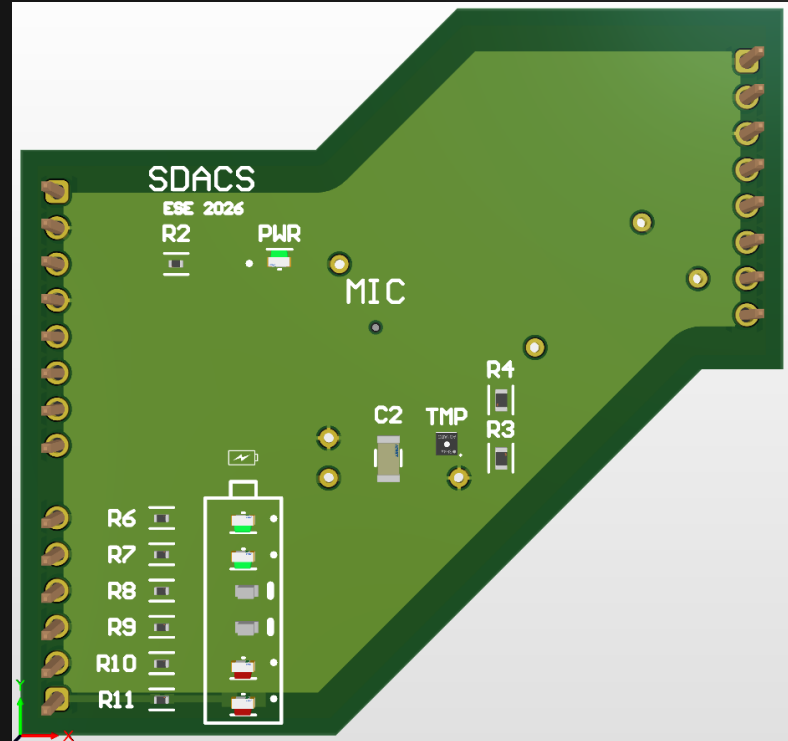
Budget Constraints,
Battery Handling and
Safety

05 RISK MITIGATION PROGRESS

Custom PCB Complexity

- Simplified PCB Design
- Sensors Board
- Compatible w/ MCU (Adafruit Metro)

*****Successfully mitigated the risk*****

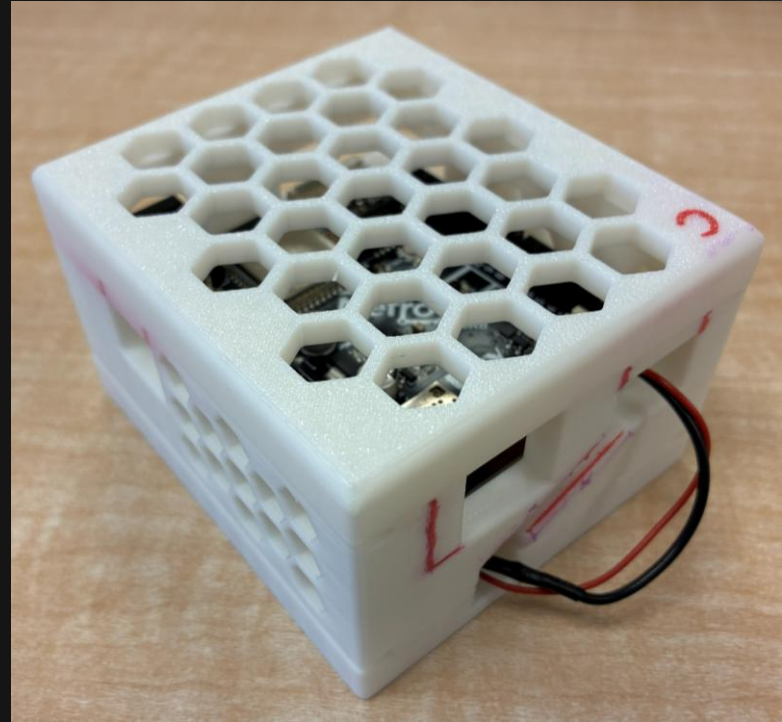


05 RISK MITIGATION PROGRESS

Battery Handling and Safety

- Enclosure designed with proper ventilation
- Air gap between battery and MCU (~1.4cm) to limit EMI and heat transfer

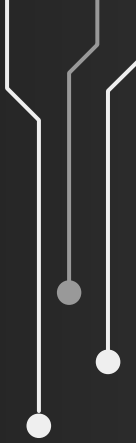
*****Successfully mitigated the risk*****



05

RISK ANALYSIS AND RISK MANAGEMENT PLAN

Risk	Likelihood (Low-Mod-High)	Impact (Low-Mod-High)
Competitive Landscape	Low	Low
Node Synchronization	Mod	High
<u>Custom PCB Complexity</u>	<u>Low</u>	<u>Low</u>
Scope Creep	Mod	Mod
Time Constraints	Mod	High
App and Website Development	Low	Mod
Budget Constraints	Low	Mod
<u>Battery Handling and Safety</u>	<u>Low</u>	<u>Low</u>



07 Q & A

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